

Validation Method Eligibility Module (VMEM)

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Authority: OOF®
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Canonical Language: English (UCL™)

Governed Space

Validation Method Eligibility

Canonical Definition

Validation Method Eligibility Module (VMEM) defines the governance framework for identifying candidate validation methods and determining whether they possess the minimum methodological characteristics necessary to enter formal suitability assessment within VALIDOS®.

It establishes the conditions under which a scientific, empirical, experimental, statistical, analytical, computational, documentary, comparative, expert, simulation-based, formal, cryptographic, adversarial, or other validation approach may be recognized as an **Eligible Validation Method** for a defined Validation Object and Validation Criteria Framework.

Operational Role

VMEM governs the entry gate into Validation Method governance.

Its role is not to determine which method should ultimately be selected.

Its role is to determine which candidate methods are sufficiently

legitimate, identifiable, applicable, and methodologically established to deserve further suitability assessment.

The governed progression is:

Validation Criteria Framework → Method Requirements → Candidate Methods → Eligibility Assessment → Eligible Validation Methods

Only after eligibility has been established should candidate methods proceed into deeper methodological suitability, selection, configuration, and integration.

Module Operational Space

VMEM governs:

- candidate method identification,
 - method recognition,
 - method eligibility,
 - methodological legitimacy,
 - method applicability,
 - method purpose alignment,
 - method definability,
 - method transparency,
 - method maturity,
 - methodological basis,
 - minimum documentation,
 - method availability,
 - method accessibility,
 - method constraints,
 - eligibility exclusions,
 - conditional eligibility,
 - eligibility uncertainty,
 - and method eligibility traceability.
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Module Function

VMEM applies after a sufficiently coherent Validation Criteria Framework exists and the methodological requirements necessary to examine it begin to be identified.

Its function is to prevent:

- arbitrary techniques entering formal validation,
 - undocumented procedures being represented as established validation methods,
 - methods without identifiable methodological foundations,
 - methods incapable of addressing the relevant type of Validation Criterion,
 - methods being selected solely because they are available,
 - experimental techniques being represented as established methods without qualification,
 - proprietary or opaque methods entering validation without sufficient methodological visibility,
 - methods being transferred between incompatible domains without examination,
 - and method selection beginning before candidate-method legitimacy has been established.
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Candidate Validation Method

A **Candidate Validation Method** is any proposed methodological approach that may potentially contribute to examination of one or more Validation Criteria.

Candidate status does not imply:

- eligibility,
- suitability,
- superiority,
- approval,
- selection,
- or validation readiness.

It merely establishes that the method has been identified for governance assessment.

This distinction prevents early methodological preference from being mistaken for formal method approval.

Method Eligibility

A candidate method may become an **Eligible Validation Method** when sufficient basis exists to establish that it can legitimately proceed to formal suitability assessment.

Eligibility may require that the method is:

- sufficiently identifiable,
- methodologically describable,
- relevant to the type of characteristic being evaluated,
- capable of producing or evaluating validation-relevant information,
- sufficiently transparent for its role to be understood,
- compatible with applicable governance constraints,
- and not subject to an exclusion that prevents legitimate use.

Eligibility represents a minimum methodological gate.

It does not establish that the method is the best or sufficient method for the validation.

Methodological Basis

An eligible method should possess an identifiable methodological basis.

That basis may originate from:

- scientific methodology,
- engineering practice,
- statistical methodology,
- formal methods,
- established technical practice,
- regulatory methodology,
- domain standards,
- validated organizational methodology,
- professional practice,
- cryptographic mechanisms,
- controlled simulation,
- empirical testing,
- or another defensible methodological foundation.

Novel methods are not automatically excluded.

However, their novelty, maturity, assumptions, and evidentiary basis must remain visible.

Method Identifiability

A validation method must be sufficiently identifiable to distinguish it from other methods.

Where applicable, method identification may include:

- method name,
- method type,
- methodological family,
- method version,
- originating authority or source,
- intended function,
- required inputs,
- expected outputs,
- and relevant methodological references.

A generic statement such as **"the system was tested"** does not establish an identifiable validation method.

Method Definability

An eligible method must be sufficiently describable for a qualified evaluator to understand what the method is intended to do.

The description should allow governance to distinguish:

- the method itself,
- its implementation,
- its configuration,
- its execution,

- and its resulting evidence.

This prevents tools, software products, or procedures from being confused with the underlying methodological approach.

Method Relevance

A method may be legitimate in general while irrelevant to a specific Validation Criterion.

VMEM therefore examines whether the candidate method has at least a plausible methodological relationship to the type of condition requiring examination.

For example:

A documentary validation method may be eligible for examining whether a required approval exists.

It may not be eligible by itself for determining physical performance under operational load.

Method legitimacy and method relevance are separate questions.

Method Applicability

Candidate methods may have applicability boundaries.

These may concern:

- object type,
- system type,
- population,
- domain,
- operational environment,
- jurisdiction,
- lifecycle stage,
- data availability,
- scale,
- complexity,
- or required Validation Depth.

VMEM requires material applicability conditions to remain visible before suitability assessment.

Method Transparency

A method must provide sufficient methodological visibility for its validation role to be governable.

Transparency does not necessarily require public disclosure of

proprietary intellectual property.

However, sufficient information must exist to understand:

- what the method evaluates,
- its general mechanism,
- material assumptions,
- material limitations,
- required inputs,
- expected outputs,
- and the basis upon which its results may be interpreted.

Where this cannot be established, eligibility may be restricted or denied.

Method Maturity

Validation methods may exist at different maturity levels.

A method may be:

- established,
- standardized,
- validated,
- emerging,
- experimental,
- prototype,
- or insufficiently characterized.

VMEM does not automatically exclude emerging or experimental methods.

Instead, maturity becomes a governed attribute capable of influencing later suitability, combination, independence, and reliance decisions.

Method Availability and Accessibility

A theoretically appropriate method may not be practically available for a particular validation.

Constraints may include:

- unavailable equipment,
- unavailable expertise,
- inaccessible datasets,
- proprietary restrictions,
- insufficient computational resources,
- unavailable reference environments,
- or legal limitations.

Availability alone does not determine methodological eligibility.

However, a method that cannot actually be accessed or implemented under the governed validation conditions may require conditional eligibility or exclusion from the usable candidate set.

Eligibility Exclusions

A candidate method may be classified as ineligible where material conditions prevent legitimate validation use.

Possible exclusions include:

- no identifiable methodological basis,
- inability to address the relevant characteristic,
- fundamental incompatibility with the Validation Object,
- prohibited use,
- unacceptable opacity,
- unresolved methodological defect,
- invalid underlying assumptions,
- obsolete methodology,
- or another material governance condition.

An exclusion must remain traceable to its basis.

Conditional Eligibility

Some methods may be eligible only under specified conditions.

Examples include:

- use only with independent corroboration,
- use only for a defined population,
- use only within a specific operating range,
- use only as a supporting method,
- use only after calibration,
- use only with specified evidence,
- or use only at a limited Validation Depth.

Conditional eligibility must remain attached to the method throughout subsequent method governance.

Eligibility Uncertainty

Where available information is insufficient to establish eligibility or ineligibility, VMEM should preserve uncertainty.

Possible states may include:

Eligible

Conditionally Eligible

Eligibility Undetermined

Ineligible

An undetermined method should not silently be treated as eligible merely because no alternative is available.

Method Diversity at Candidate Stage

VMEM should avoid prematurely narrowing candidate methods to one familiar methodological family.

Where relevant, candidate identification may consider different approaches capable of examining the same Validation Criterion.

This creates the foundation for later methodological comparison and triangulation.

However, candidate diversity does not require every conceivable method to be included.

The objective is meaningful methodological coverage, not uncontrolled method accumulation.

Minimum Implementation Framework

1. Identify Method Requirements

Determine the broad methodological capabilities required by the Validation Criteria Framework.

2. Identify Candidate Methods

Identify potentially relevant validation approaches without treating candidate status as approval.

3. Establish Method Identity

Record sufficient information to distinguish and describe each candidate method.

4. Assess Minimum Eligibility

Determine whether the method possesses:

- an identifiable methodological basis,
 - plausible relevance,
 - applicable operating conditions,
 - sufficient transparency,
 - and no fundamental exclusion.
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5. Establish Eligibility Status

Classify the method as:

- Eligible,
- Conditionally Eligible,
- Eligibility Undetermined,
- or Ineligible.

6. Preserve Eligibility Traceability

Maintain:

- Candidate Method Identifier,
- method type,
- methodological basis,
- intended validation role,
- relevant Validation Criteria,
- applicability,
- maturity,
- transparency conditions,
- constraints,
- exclusions,
- eligibility status,
- conditions,
- version,
- and sufficient lifecycle traceability.

Eligibility Is Not Suitability

VMEM establishes a strict distinction:

Eligible Validation Method ≠ Suitable Validation Method

Eligibility asks:

Can this method legitimately enter consideration?

Suitability asks:

Is this method appropriate for this specific Validation Object, criterion, depth, context, and intended determination?

A method may therefore be eligible but later rejected during suitability assessment.

This separation prevents minimum legitimacy from being confused with validation fitness.

Eligibility Change Governance

Method eligibility may change because of:

- new scientific evidence,
- discovered methodological defects,
- regulatory change,
- changed applicability,
- new security or integrity concerns,
- improved methodological understanding,
- method obsolescence,
- or changes to the Validation Object or Validation Criteria.

Material eligibility changes must remain documented and propagated into subsequent method governance.

Use Case 1 — AI Model Validation

Scenario

An organization identifies several possible methods for validating robustness of an AI model, including benchmark testing, adversarial testing, expert review, simulation, and an experimental automated evaluation technique.

Application

VMEM does not immediately select the preferred method.

Each candidate is examined for methodological basis, relevance, applicability, maturity, transparency, and material exclusions.

The experimental technique is classified as **Conditionally Eligible** because its methodological maturity remains limited.

Result

VALIDOS® retains methodological innovation without allowing experimental status to be mistaken for established validation fitness.

Use Case 2 — Operational Infrastructure

Scenario

A validation team proposes a familiar documentary review method for assessing the operational resilience of a critical infrastructure component.

Application

The documentary method is methodologically legitimate but cannot by itself examine the component's actual behavior under

operational stress.

Result

VMEM may recognize the method as eligible for documentary criteria while excluding it from candidate coverage of operational-performance criteria.

Method eligibility remains tied to the validation role it can legitimately perform.

Canonical Closing Statement

Validation Method Eligibility Module establishes the canonical entry gate through which candidate validation methods become eligible for formal methodological consideration within VALIDOS®. By governing method identification, methodological basis, relevance, applicability, transparency, maturity, constraints, exclusions, conditional eligibility, uncertainty, and traceability, VMEM ensures that subsequent method selection begins from a governed set of legitimate candidates rather than arbitrary techniques, habitual practices, or methods chosen primarily because they are available.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Next module: Validation Method Suitability Module \(VMSM\)](#)

Related Documents

[→ Validation Method Standard](#)

[→ About Validation Method Standard](#)

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